

Cyberattack Article Summarization & Translation Bot

Project Overview:

This project is a full-stack application that automatically collects, summarizes, and displays the latest cybersecurity-related articles. It uses HTML, CSS, and JavaScript for the user interface, Node.js for the backend, and Firebase Firestore as the database.

Key Features:

- Automatically scrapes articles related to cyberattacks from RSS feeds (e.g., Google News, BBC, TechCrunch, ZDNet, The Verge).
- Every 12 hours, it searches for new articles using a scheduled cron job.
- For each article, it uses OpenAI's GPT-4 API to generate:
 1. A 4-5 sentence summary
 2. Type of cyberattack (phishing, ransomware, etc.)
 3. Severity level (Low, Medium, High, Critical)
 4. Rarity (Common, Rare, Zero-Day, Targeted)
 5. How to prevent such attacks
 6. Key indicators of the attack
- Results are stored in Firebase in English only.
- Users can view collected articles and translate them to other languages in the UI (e.g., Hebrew, Hindi, Tamil, etc.) using on-the-fly translation.
- Old articles are not reprocessed; only new articles are collected from RSS feeds since project launch.

Technologies Used:

- Frontend: HTML, CSS, JavaScript
- Backend: Node.js, Axios, dotenv, node-cron

- ML Integration: OpenAI GPT-4 API
- Database: Firebase Firestore
- Scheduler: node-cron

Purpose:

The goal is to automate real-time monitoring and summarization of cyberattack news, making the information digestible and available in multiple languages through a single interface.